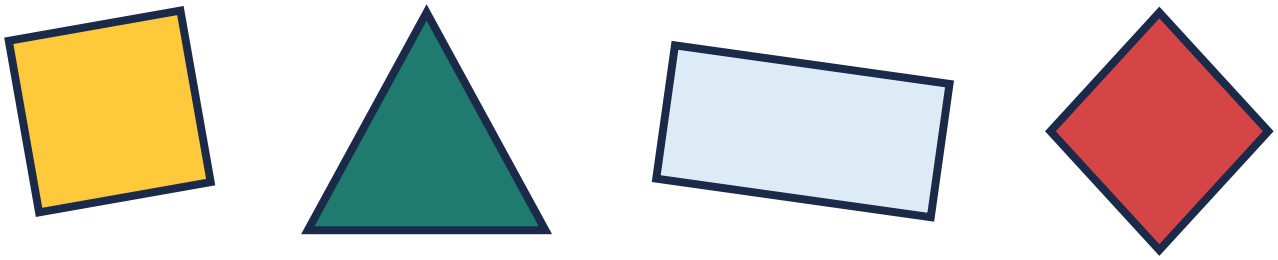
FOR PARENTS • PRIMARY 6, AFTER THE PSLE

Weekly Dose of Thinking Math

One dose a week for five weeks: short sets of questions for every Math Inquirers lesson, a tip for you, and the answers at the back.



Notice it. Test it. Explain why.

Math Inquirers Holiday Program
Preciouslyours Pte Ltd

BEFORE YOU START

00 FOR GROWN-UPS How to help

1. **Ask before you tell.** Start with "What do you notice?" and let your child answer first.
2. **Use real things.** Paper, a pencil and, for coding lessons, a computer with Scratch (free at scratch.mit.edu). Let your child show you the program and explain what each block does.
3. **Give hints in three steps.** Look again. Try it with objects. Try a smaller example. Say the answer last.
4. **Ask "How do you know?"** even when the answer is right. The reason matters as much as the answer.
5. **Praise the thinking.** "I like how you checked that" works better than "You are clever".
6. **Keep it short.** Ten minutes and two or three questions is enough. Stop while it is still fun.
7. **Answers are at the back.** Look only after your child has tried.

> **Hard question? Great. Getting stuck is where the thinking happens.**

One dose a week

Each week has two lessons and about 18 questions. Do two or three a day, Monday to Friday, and keep the weekend for the Challenge questions if your child enjoys them.

The three levels

Core is for everyone. **Think Harder** stretches the idea. **Challenge** is for strong thinkers, and it is fine to leave it for another day.

What your child is doing in class

Ten 90-minute live online lessons over five weeks, in small groups. Each lesson starts with a hands-on puzzle, then a short teaching part, then the workbook. These pages practise the same ideas at home.

W1 Code the Pattern

LESSON 1 • SEQUENCES AND THE NTH TERM

Core

1 Write the next two terms: 8, 13, 18, 23, ...

2 Write the rule for the nth term of 3, 5, 7, 9, ...

3 Find the 10th term of 7, 10, 13, 16, ...

4 Write the next two terms: 40, 36, 32, 28, ...

Think Harder

5 Find the 100th term of 12, 19, 26, 33, ...

6 Is 200 a term of 2, 9, 16, 23, ...? Explain.

7 Matchstick houses in a row: 1 house uses 6 sticks, 2 houses use 11, 3 houses use 16. How many houses can you make with 100 sticks?

What your child learns

Finding the rule of a number pattern, testing it in a short program, and finding the 100th term without listing them all.

Tip for you

Make a matchstick pattern of triangles or squares. Ask how many sticks the 10th figure needs before you build it.

Ask

"Can you test your rule on the next figure?"

Challenge

8 Pattern 1 is 1 tile, pattern 2 is a 2 by 2 square of tiles, pattern 3 is a 3 by 3 square. How many tiles are in pattern 10? How many more tiles does pattern 11 need than pattern 10?

9 A straight fence is 60 m long. There is a post every 4 m, including one at each end. How many posts are there?

W1 Prime Hunters

LESSON 2 • PRIMES, FACTORS AND PRIME FACTORISATION

Core

1 List all the factors of 18.

2 Is 37 prime? How do you know?

3 What is 29 mod 4 (the remainder when 29 is divided by 4)?

4 Write 30 as a product of primes.

Think Harder

5 How many factors does 64 have? List them.

6 Find the smallest number that has exactly 6 factors.

7 Write 100 as the sum of two primes in every possible way.

What your child learns

Prime numbers, factors, factor trees, and a program that counts factors.

Tip for you

Pick a number under 100 and ask whether it is prime. Let your child test it one divisor at a time.

Ask

"What would you try next if it is not divisible by 2, 3 or 5?"

Challenge

8 What is the largest prime number below 200?

9 Ali multiplies two different primes and gets 91. Mei multiplies three different primes and gets 105. Which primes did each of them use?

W2 LESSON 3 • NEGATIVE NUMBERS Below Zero

Core

1 Put in order, smallest first: −4, 2, −10, 0, −1.

2 It is −3 °C. The temperature rises by 8 degrees. What is it now?

3 It is 2 °C. The temperature falls by 7 degrees. What is it now?

4 Which is colder, −6 °C or −11 °C? By how many degrees?

Think Harder

5 Work out −7 + 12 and 5 − 13.

6 A diver is at −14 m. A seagull is flying at 9 m. How far apart are they?

7 Work out 4 − (−6). Hint: how far is it from −6 to 4?

What your child learns

Negative numbers on a number line, temperatures and floors below zero, and programs that go up and down.

Tip for you

Look at the weather in a cold city and ask for the difference between its low and ours.

Ask

"Which way do you move on the number line?"

Challenge

8 Put −4, −3, −2, −1, 0, 1, 2, 3, 4 in a 3 by 3 grid so that every row, every column and both diagonals add up to 0.

9 At midnight it was −6 °C. It rose 2 degrees every hour until 7 a.m., then 3 degrees every hour. What was the temperature at 10 a.m.?

W2 Algebra Machines

LESSON 4 • EXPRESSIONS AND EQUATIONS

Core

1 The rule is OUT = 4n + 1. What is OUT when n = 6?

2 IN 1, 2, 3, 4 gives OUT 3, 5, 7, 9. Write the rule with n.

3 Solve 3n = 54.

4 A machine multiplies by 6, then takes away 4. OUT is 38. What was IN?

Think Harder

5 IN 1, 2, 3, 4 gives OUT 1, 7, 13, 19. Find the rule. What is OUT when IN is 20?

6 Solve 9n − 7 = 65.

7 Machine A does × 2, then + 10. Machine B does + 5, then × 2. Put 7 into both. What do you notice? Explain with n.

What your child learns

Number machines and simple algebra: finding the rule, working backwards, and a program that repeats until it hits a target.

Tip for you

Say a number-machine riddle: "I think of a number, double it, add 3 and get 17." Swap and let your child make one up.

Ask

"What is the opposite of each step?"

Challenge

8 A machine does × 3, then − 4. Which IN number comes out as the SAME number?

9 Two machines are joined: the OUT of the first goes into the second. Each one does × 2, then + 1. IN 5 gives OUT 23. Which IN gives OUT 63?

W3 LESSON 5 • COUNTING Every Possible Way

Core

1 A shop sells 3 shirts and 5 pairs of trousers. How many different outfits of 1 shirt and 1 pair of trousers can you make?

2 List every 2-digit number you can make from the cards 4, 6 and 9, using each card at most once.

3 4 friends meet. Every pair shakes hands once. How many handshakes?

4 How many times does this say “hi”? repeat 4 { repeat 6 { say hi } }

Think Harder

5 A set lunch is 1 main, 1 drink and 1 dessert. There are 4 mains and 3 drinks. There are 60 different set lunches. How many desserts are there?

6 How many 3-digit numbers can you make from the cards 1, 2, 3 and 4, using each card at most once? How many of them are even?

What your child learns

Counting every possible way: tree diagrams, the multiplication rule, handshakes, and nested loops.

Tip for you

Ask how many outfits can be made from 3 tops and 4 skirts, then draw all of them.

Ask

"How do you know you have not missed or counted twice?"

7 In a league of 6 teams, every team plays every other team twice (once at home, once away). How many matches are played?

Challenge

8 How many different triangles can you make by joining 3 of the 5 corners of a pentagon?

9 How many 2-digit numbers have two different digits?

W3 LESSON 6 • PROBABILITY
Fair or Unfair?

Core

1 Roll a dice. What is the probability of rolling a 4?

2 Roll a dice. What is the probability of rolling an even number?

3 A bag has 3 red and 7 yellow counters. You pick one without looking. What is the probability it is yellow?

4 You flip a coin 200 times. About how many heads do you expect?

Think Harder

5 A spinner has 10 equal parts numbered 1 to 10. What is the probability of spinning a prime number?

6 Roll two dice and add. What is the probability that the total is 9?

What your child learns

Probability as a fraction, fair and unfair games, and a program that rolls a die 1000 times.

Tip for you

Play a dice game with a friend. Ask whether it is fair, and then keep a tally of 50 throws.

Ask

"Why is it close, but not exactly what we expected?"

7 A bag has 5 red counters and some blue counters. The probability of picking red is 1/4. How many blue counters are there?

Challenge

8 A bag has red and blue counters in the ratio 2 : 3. What is the probability of picking red?

9 A number from 1 to 30 is picked at random. What is the probability that it is a multiple of 3 or a multiple of 5?

W4 LESSON 7 • ANGLES IN POLYGONS
Turtle Geometry

Core

1 A turtle draws a regular hexagon. How many degrees does it turn at each corner?

2 What is each inside angle of a regular hexagon?

3 Fill in the turn so the turtle draws a square: repeat 4 [move 100, turn right ___]

4 A regular polygon has a turn angle of 72°. How many sides does it have?

Think Harder

5 Each inside angle of a regular polygon is 4 times its turn angle. How many sides does the polygon have?

6 Each inside angle of a regular polygon is 140°. What do all its inside angles add up to?

What your child learns

Turtle geometry: turning angles, regular polygons and stars, and total turning.

Tip for you

Walk a regular shape on the floor: five steps, turn, repeat. How much did you turn in total when you came back?

Ask

"Why does the turtle always turn 360 degrees in total?"

7 Mei's turtle draws a shape with 4 sides. Its first three turns are 70°, 100° and 90°. It comes home facing the same way after going round once. What is its last turn? What is the inside angle at that corner?

Challenge

8 Ali wants a regular pentagon. He types the inside angle by mistake: repeat 5 [move 100, turn right 108]. Does his shape close? How many repeats would make it close, and what would it draw?

9 How many different stars with 9 points can a turtle draw using equal moves and equal turns? Give the turns.

W4 LESSON 8 • CLOCK (MODULAR) ARITHMETIC
Secret Codes

Core

1 It is 10 o'clock. What time will it be 5 hours later?

2 Work out 20 mod 12.

3 What is the remainder when 45 is divided by 7?

4 Today is Monday. What day will it be 10 days from today?

Think Harder

5 The pattern CODECODECODE... goes on and on. What is the 100th letter?

6 It is 10 o'clock now. What time will it be 1000 hours from now (on a 12-hour clock)?

7 Find the smallest number bigger than 10 that leaves a remainder of 3 when it is divided by 4 and also when it is divided by 5.

What your child learns

Secret codes: clock arithmetic, shift codes, cracking a code, and a program that scrambles a message.

Tip for you

Send each other a short message in a shift code and crack it by looking for the most common letters.

Ask

"What happens when you go past Z?"

Challenge

8 Mei makes a new code: multiply each letter number by 2, then take mod 26. Why can her friend not always decode it?

9 Ben's code multiplies each letter number by 3, then takes mod 26. Code the word CAB. Do two letters ever get the same code letter?

W5 Data Detectives

LESSON 9 • AVERAGES AND DATA

Core

1 Find the mean of 3, 5, 7 and 9.

2 Find the mean of 10, 0, 8 and 6.

3 The mean of 5 numbers is 11. What is their total?

4 Four bags have a mean mass of 2.5 kg. What is their total mass?

Think Harder

5 The mean height of 4 children is 150 cm. A fifth child joins the group and the mean height becomes 152 cm. How tall is the fifth child?

6 Mei's mean score for 3 tests is 84. What must she score in the 4th test to raise her mean to 86?

7 Write 5 whole numbers with a mean of 6 and a mode of 4.

What your child learns

Data: mean, median and mode, what changes the mean, and a program that finds the mean of a list.

Tip for you

Collect the heights or pocket money of your family and work out the mean and the median. Which describes them better?

Ask

"When is the mean misleading?"

Challenge

8 Five numbers have a mean of 8. One number is removed, and the mean of the other four is 7. Which number was removed?

9 A basketball player's mean score for 9 games is 30 points. How many points must she score in the 10th game to raise her mean to 32?

W5 LESSON 10 • MIXED: MATHS AND CODE
Showcase Challenge

Core

1 Find the 20th term of 5, 8, 11, 14, ...

2 Write 60 as a product of prime numbers.

3 The temperature is −3 °C. It falls by 5 °C. What is it now?

4 A machine does × 4, then − 6. What comes out when 5 goes in?

Think Harder

5 A row of squares uses 4, 7, 10, 13, ... matchsticks. Which pattern number uses exactly 100 matchsticks?

6 Find all the prime numbers between 80 and 100.

7 A regular polygon has an angle of 150° inside each corner. How many sides does it have?

What your child learns

Showcase: a mixed challenge round, then planning, building and testing a small puzzle program.

Tip for you

Ask your child to explain their program to you, block by block. Then try to break it with an unusual input.

Ask

"How did you test it? What would you improve?"

Challenge

8 How many of the numbers from 1 to 100 are NOT multiples of 2 or of 3?

9 A machine does × 3, then − 8. Mei puts a number in and gets the SAME number out. What is it? What if the machine does × 3, then + 8?

LOOK AFTER YOUR CHILD HAS TRIED

ALL LESSONS

Answers

Week 1 • Lesson 1: Code the Pattern

1. 28, 33 (add 5)
2. 2n + 1
3. 34 (rule 3n + 4: 3 × 10 + 4)
4. 24, 20 (take away 4)
5. 705. The rule is 7n + 5, and 7 × 100 + 5 = 705.
6. No. The rule is 7n − 5. The 29th term is 198 and the 30th term is 205, so 200 is skipped.
7. 19 houses. The rule is 5n + 1: 19 houses use 96 sticks (4 left over) and 20 houses would need 101.
8. 100 tiles. Pattern 11 has 121 tiles, so 21 more. The differences are 3, 5, 7, 9, ..., so this is NOT a constant-difference sequence.
9. 16 posts. 60 ÷ 4 = 15 gaps, and there is one more post than gaps.

Week 1 • Lesson 2: Prime Hunters

1. 1, 2, 3, 6, 9, 18
2. Yes. Its only factors are 1 and 37 (2, 3, 5 do not divide it, and 7 × 7 = 49 is already bigger than 37).
3. 1 (4 × 7 = 28)
4. 2 × 3 × 5
5. 7: 1, 2, 4, 8, 16, 32, 64 (8 × 8 counts once).
6. 12 (1, 2, 3, 4, 6, 12). Every number below 12 has fewer than 6 factors.
7. 6 ways: 3 + 97, 11 + 89, 17 + 83, 29 + 71, 41 + 59, 47 + 53.
8. 199. Test 2, 3, 5, 7, 11 and 13 (14 × 14 = 196, 15 × 15 = 225): none of them divides 199.
9. Ali: 7 × 13. Mei: 3 × 5 × 7.

Week 2 • Lesson 3: Below Zero

1. −10, −4, −1, 0, 2
2. 5 °C
3. −5 °C
4. −11 °C, by 5 degrees
5. 5 and −8
6. 23 m (14 up to sea level, then 9 more)
7. 10
8. One answer: top row −3, 2, 1; middle row 4, 0, −4; bottom row −1, −2, 3. 0 must be in the middle and 4 and −4 cannot be in corners; the other 7 answers are turns and flips of this one.
9. 17 °C (−6 + 7 × 2 = 8 at 7 a.m., then 8 + 3 × 3 = 17)

Week 2 • Lesson 4: Algebra Machines

1. 25
2. OUT = 2n + 1
3. n = 18
4. 7 (38 + 4 = 42, 42 ÷ 6 = 7)
5. OUT = 6n − 5; IN 20 gives 115.
6. n = 8 (65 + 7 = 72, 72 ÷ 9 = 8)
7. Both give 24. (n + 5) × 2 = 2n + 10, so they always give the same OUT.
8. 2 (2 × 3 − 4 = 2; 3n − 4 = n)
9. 15 (63 → 31 → 15; check 15 → 31 → 63)

Week 3 • Lesson 5: Every Possible Way

1. 15 (3 × 5)
2. 6 numbers: 46, 49, 64, 69, 94, 96
3. 6 (3 + 2 + 1)
4. 24 (4 × 6)
5. 5 (4 × 3 = 12 and 60 ÷ 12 = 5)
6. 24 (4 × 3 × 2); 12 are even (the last card is 2 or 4: 2 choices, then 3 × 2 for the others)
7. 30 (6 × 5: each ordered pair is one home match)
8. 10
9. 81 (90 two-digit numbers, minus the 9 with both digits the same: 11, 22, ..., 99)

Week 3 • Lesson 6: Fair or Unfair?

1. 1/6
2. 3/6 = 1/2 (2, 4 or 6)
3. 7/10
4. About 100 (1/2 × 200), probably not exactly 100
5. 4/10 = 2/5 (2, 3, 5, 7)
6. 4/36 = 1/9 (3+6, 4+5, 5+4, 6+3)
7. 15 (5 out of 20 is 1/4)
8. 2/5
9. 14/30 = 7/15 (10 multiples of 3, 6 of 5, and 15 and 30 are counted twice: 10 + 6 − 2 = 14)

Week 4 • Lesson 7: Turtle Geometry

1. 60° (360 ÷ 6)
2. 120° (180 − 60)
3. 90
4. 5 (360 ÷ 72): a regular pentagon
5. 10 (turn + 4 × turn = 180°, so the turn is 36° and 360 ÷ 36 = 10; check: inside 144° = 4 × 36°)
6. 1260° (turn 40°, so 9 sides; 9 × 140° = 1260°; check (9 − 2) × 180° = 1260°)
7. Last turn 100° (360 − 70 − 100 − 90); inside angle 80° (180 − 100)
8. No: 5 × 108° = 540° is not a whole number of turns. 108 × 10 = 1080 = 3 × 360, so 10 repeats close it: a 10-point star that goes round 3 times (each point 72°).
9. 2: turn 80° (round 2 times) and turn 160° (round 4 times). Turn 120° (3 times round) only draws a triangle 3 times; turns of 200°, 280° ... give the same stars turned the other way.

Week 4 • Lesson 8: Secret Codes

1. 3 o'clock (15 − 12 = 3)
2. 8 (20 = 1 × 12 + 8)
3. 3 (45 = 6 × 7 + 3)
4. Thursday (10 mod 7 = 3; Monday + 3 days)
5. E (100 mod 4 = 0, so the 100th letter is the 4th letter of CODE)
6. 2 o'clock (1000 mod 12 = 4, and 10 + 4 = 14, 14 − 12 = 2)
7. 23 (3 more than a multiple of 20: 23 = 5 × 4 + 3 = 4 × 5 + 3)
8. Two letters can give the same code letter: A (0) and N (13) both give 0 = A (13 × 2 = 26). So every code letter could stand for two different letters.
9. GAD (C = 2 gives 6 = G, A = 0 gives 0 = A, B = 1 gives 3 = D). No: all 26 letters get different code letters, because 3 and 26 have no common factor.

Week 5 • Lesson 9: Data Detectives

1. 6 (24 ÷ 4)
2. 6 (24 ÷ 4: the 0 counts as a value)
3. 55
4. 10 kg
5. 160 cm (5 × 152 = 760, 4 × 150 = 600, 760 − 600 = 160)
6. 92 (4 × 86 = 344, 3 × 84 = 252, 344 − 252 = 92)
7. Open, e.g. 4, 4, 5, 8, 9 (total 30, and 4 appears most often). Check the total is 30.
8. 12 (5 × 8 = 40, 4 × 7 = 28, 40 − 28 = 12)
9. 50 (10 × 32 = 320, 9 × 30 = 270, 320 − 270 = 50)

Week 5 • Lesson 10: Showcase Challenge

1. 62 (nth term 3n + 2)
2. 2 × 2 × 3 × 5
3. −8 °C
4. 14
5. Pattern 33 (3n + 1 = 100)
6. 83, 89 and 97
7. 12 (the turn at each corner is 180° − 150° = 30°, and 360 ÷ 30 = 12)
8. 33 (multiples of 2: 50, of 3: 33, of both (6): 16; 100 − (50 + 33 − 16) = 33)
9. 4 (4 × 3 − 8 = 4). With + 8 it is −4 (−4 × 3 + 8 = −4).